Order-4222 cubic descent: an explicit witness in an unresolved isogeny class

Result and claim boundary

This folder gives a reproducible, end-to-end GHS transfer for one source elliptic curve of order `4222 = 2*2111` over `F_(2^12)`. The descended curve is a smooth nonhyperelliptic genus-three curve over `F_16`. Its Jacobian lies in LMFDB isogeny class `3.16.a_a_ev`, whose `has_jacobian` value was `0` (**unknown**) in both the 2026-10-19 local snapshot and the live SQL mirror queried on 2026-10-28. The explicit quartic and its L-polynomial constitute a positive class witness suitable for catalog review. The source equation is new to this audit; global publication novelty remains unverified.

This curve belongs to the previously identified nonsplit quadratic-pole family. The construction is neither a newly established geometric family nor a new attack method. The completed dynamic Magma transfer and attack took about 271 times the mean cost of a matched source-group rho solve. Thus this particular curve is **not** a measured GHS speedup.

Exact source and cover

Let `L = F_2[z]/(z^24+z^16+z^15+z^14+z^13+z^10+z^9+z^7+z^5+z^3+1)`. The unique subfields `k` and `K` of `L` have 16 and 4096 elements. All decimal elements below are 24-bit coefficient masks in the basis `1,z,...,z^23`, rather than integer coefficients. The source is

```
E/K: Y^2 + XY = X^3 + 5522909 X^2 + 1610778
#E(K) = 4222 = 2*2111,  2111 prime
G = (1610779, 12717539),  order(G) = 2111
Q = (10751255, 15683006) = 7G
```

Set `q=16`, `t=16191718`, and `A=5857459` in `L`. Then `t+t^q=1`, `Tr_(L/F_256)(A)=0`, and `(A*A^(q^3))^2=1610778`. The three conjugate Artin-Schreier functions are

```
f_i(x) = A^(q^i)/(x+t^(q^i))
       + A^(q^(i+3))/(x+t^(q^(i+3))),  i=0,1,2.
```

They lie in `K(x)`, satisfy `f_0+f_1+f_2=0`, and descend cyclically under q-Frobenius. Their compositum gives a V4 cover. An exact affine model of the descended curve over `k` is

```
u^4 + 4534198 u =
  (5763596 + x + 4626534 x^2 + 5763597 x^3)
  / (5522908 + x^2 + x^4).
```

The exported smooth projective plane quartic and its affine-to-canonical map are in `canonical_4222.json`. Magma 2.30-1 checked geometric genus 3, nonhyperellipticity, smoothness, and

```
L_C(T) = 1 + 125 T^3 + 4096 T^6.
```

This polynomial is the exact `poly` value `[1,0,0,125,0,0,4096]` for LMFDB class `3.16.a_a_ev`. The class has `p_rank=3`, `is_simple=true`, and `is_geometrically_simple=false`. The pinned polynomial and canonical-map receipt is `canonical_4222_receipt.json`. An independent Sage count finds `#C(F_16)=17`, `#C(F_256)=257`, and `#C(F_4096)=4472`; the last count uses the 1117 split affine fibers of the V4 cover plus four points at infinity. Newton identities and the genus-three functional equation recover the same L-polynomial. See `independent_counts_4222.json`.

Subgroup-preserving correspondence

Magma constructed the genus-one quotient `C_0/K` defined by `y_0^2+y_0=f_0(x)` and an explicit isomorphism from its elliptic model to `E`. The rational map coefficients are in `quotient_map_4222.json`; the exact Magma input and output are in `quotient_4222.magma` and `quotient_4222_receipt.json`. Three split fibers above `R=2G`, `R+G=3G`, and `R+Q=9G` are in `preimages_4222.json`.

For the quotient map `pi:C_K -> E`, define

```
Phi(P) = sum_(i=0)^2 Frob_q^i( pi^*([P+R]-[R]) ).
```

This is a homomorphism from the source elliptic group to `J(C)(k)`. In the three `V4` quotient components, push-pull gives `pi_i,* pi_i^* = [2]` and the cross terms vanish. Projecting `Phi` back to the first quotient after base change to `K` gives `[2]P`. Therefore its kernel has no odd torsion; in particular, it is injective on the prime-order 2111 subgroup. The executable Sage Jacobian calculation independently verified

```
Phi(G) != 0,   2111 Phi(G) = 0,
Phi(Q) != 0,   2111 Phi(Q) = 0,
Phi(Q) = 7 Phi(G).
```

The map receipt includes the exact quartic fibers and five Boolean checks. `Order4222Gate.lean` kernel-checks the finite count, source cofactor, scalar, and odd-subgroup doubling arithmetic. It does **not** formalize finite-field descent, smoothness, the quotient correspondence, or the Jacobian identities; those claims have Sage and Magma evidence and a mathematical argument here.

Complete attack versus matched rho

For a stricter same-backend comparison, `full_attack_4222.magma` starts with the exact source points, rebuilds the quotient and split fibers, evaluates their canonical-quartic images, forms the traced divisors, lifts them to `F_256`, and runs the line sieve. Its checked job verified the 2111-order and `7` relation both before and after lifting. The separate timed job omitted the expensive principal-divisor assertions, while recovering and verifying `7` through the same dynamic path. It measured 0.190 CPU seconds through source mapping, 0.250 CPU seconds through all pre-index work, 0.210 CPU seconds in index calculus, and 0.460 CPU seconds from start to verified source answer (0.550 server seconds). The canonical quartic equation is reusable curve setup supplied as an exact input to both jobs.

An explicit Floyd-rho implementation on the same `E,G,Q,2111` in Magma 2.30-1 completed 100/100 seeded runs, with median 207 group-step calls and 0.170 CPU seconds for all 100, or 0.00170 CPU seconds per solve. The 0.460-second dynamic Magma attack was about 271 times that matched rho mean. The older precomputed-divisor attack core took 0.410 CPU seconds, 241 times rho. Local Sage Jacobian setup took 13.65 wall seconds and two map evaluations took 5.33 wall seconds; these are separate-machine measurements and are excluded from the Magma CPU ratio. The full dynamic Magma run supersedes them for the main same-backend comparison. The two failed preliminary dynamic jobs are retained with `failed_syntax` and `failed_verbose` names; neither is counted as a completed solve. The exact inputs and receipts are summarized in `cost_4222.json`.

Bounds and prior work

For the normalized ordinary models `Y^2+XY=X^3+aX^2+b` over `F_(q^3)` with `q=2^m`, the nonsplit quadratic-pole construction has the exact coefficient criterion

```
Tr_(F_(q^3)/F_2)(a)=1,
sqrt(b) in { A*A^(q^3) : 0 != A in ker Tr_(F_(q^6)/F_(q^2)) }.
```

The norm image has exactly `(q^3+q-2)/2` values. The count of models added outside the triangle screen is `(q^3-q^2+4q-4)/4` when `m` is even, and `(q^3-q^2-4)/4` when `m` is odd. Together with triangle and Hess screens, the union contains `3q^3/2+q-2` or `3q^3/2-2` of the `2(q^3-1)` normalized models, respectively. At `q=16`, these numbers are 975 additional nonsplit models and 6158/8190 in the three-screen union. The remaining 2032 models are simply outside these screens; the count gives no resistance bound against other covers or isogenies. A screen count also does not imply a concrete attack advantage. See the companion nonsplit-family report and executable population count for the all-q proof and the `q=2,4,8,16` enumeration.

Momose and Chao (2006) and Murai et al. (2022) classify related binary cover routes, so the geometry cannot be called a new family from these data. Joux and Vitse (2012) studied cover-and-decomposition attacks and isogeny walks. Searches for the exact class label, Weil polynomial, source order, and subgroup prime did not find an indexed publication of this exact instance; those negative searches do not prove global novelty. The per-source prior-work ledger is `prior_work_4222.json`. The class's `has_jacobian=0` is an uncertainty marker, not a negative theorem, per the LMFDB inventory's three-valued Jacobian status.

Primary links: Momose-Chao 2006, Murai et al. 2022 abstract, Joux-Vitse 2012, LMFDB inventory, LMFDB class.